

**National College of Ireland  
Project Submission Sheet**

**Programme:** MSCAIJAN25I, MSCAIBUSJAN25I **Year:** 2025

**Module:** Foundations of Artificial Intelligence

**Lecturer:** Sallar Khan

**Submission Due Date:** Monday, 14<sup>th</sup> April 2025 @12:00 A.M

**Project Title:** AI Powered Application to serve GPs and patient in Ireland

**Word Count** 2000

**Team Members:** 4 Members

| Student Name                               | Student No | Mail Id                    | Programme     |
|--------------------------------------------|------------|----------------------------|---------------|
| Manikanta Muniswamy Sampangiramaiah        | x23373458  | x23373458@student.ncirl.ie | MSCAIBUSJAN25 |
| Dheeraj Gopalkrishna                       | x24135739  | x24135739@student.ncirl.ie | MSCAIJAN25I   |
| Swapnil Nandkishor Suryawanshi             | x24126608  | x24126608@student.ncirl.ie | MSCAIBUSJAN25 |
| Madhava Lakshmi Satya Mohan Samanthakurthi | x24123684  | x24123684@student.ncirl.ie | MSCAIBUSJAN25 |

**I hereby certify that the information contained in this (my submission) is information pertaining to research I conducted for this project. All information other than my own contribution will be fully referenced and listed in the relevant bibliography section at the rear of the project.**

**ALL internet material must be referenced in the references section. Students are encouraged to use the Harvard Referencing Standard supplied by the library. To use other author's written or electronic work is illegal (plagiarism) and may result in disciplinary action. Students may be required to undergo a viva (oral examination) if there is suspicion about the validity of their submitted work.**

**Team Agreement**

We confirm that the information provided above is accurate and each member has contributed fairly to the work.

| Student Name                               | Signature    | Date     |
|--------------------------------------------|--------------|----------|
| Manikanta Muniswamy Sampangiramaiah        | Manikanta MS | April 13 |
| Dheeraj Gopalkrishna                       | Dheeraj G    | April 13 |
| Swapnil Nandkishor Suryawanshi             | Swapnil      | April 13 |
| Madhava Lakshmi Satya Mohan Samanthakurthi | Madhava      | April 13 |

# AI Powered Application to Serve GPs and Patient in Ireland

Manikanta Muniswamy Sampangiramaiah  
National College of Ireland  
Dublin, Ireland  
x23373458@student.ncirl.ie

Dheeraj Gopalkrishna  
National College of Ireland  
Dublin, Ireland  
x24135739@student.ncirl.ie

Swapnil Nandkishor Suryawanshi  
National College of Ireland  
Dublin, Ireland  
x24126608@student.ncirl.ie

Madhava Lakshmi Satya Mohan Samanthakurthi  
National College of Ireland  
Dublin, Ireland  
x24123684@student.ncirl.ie

**Abstract**—The health system in Ireland is facing challenges in its regular General Practitioner (GP) appointment systems due to their manual processes and this results in inefficiencies such as scheduling conflicts, no-shows, and administrative burdens. This project is providing an innovative idea to integrate Artificial Intelligence along with Machine Learning, Natural Language Processing, and geolocation-based services to streamline booking, rescheduling, and cancellation processes into the GP consultation system. To resolve the issues an intelligent chatbot is trained on HSE corpus to answer the medical queries, supervised learning methods like XGBoost and Random Forest is used to create no-show prediction model and an EIR code-based search module for specialist doctors. Evaluation result showcased that XGBoost performed well with improved operational efficiency and enhanced patient experience thus providing a scalable and patient-centric solution for modernizing Ireland's GP services.

**Keywords**—GP appointment system, artificial Intelligence, XGBoost, chatbot.

## I. INTRODUCTION

The Healthcare industry in Ireland is going through huge pressure to bring innovation in General Practitioner (GP) service for public and private patient care. Health segment is still following the old-fashioned way to provide the GP services and this old way of operating, results in delays, conflict with appointments and patient no show up. In few cases GPs operate with receptionist for scheduling the appointment through email, call, patient walk-in's for GP availability, which result in administrative bottle neck for actual patient who need assistance [1]. Patients who submit an medical or GP card ideally get free GP visits whereas without cards they must pay privately and may face limited availability or higher price.

As a project team, the motivation is to explore the application of AI and recent studies suggest that patient satisfaction and GP clinical outcomes are closely tied to the efficiency of appointment systems. This area is considered as urgent, making an innovation using Artificial Intelligence(AI) and particularly through Machine Learning (ML) and Natural Language Processing(NLP) which provides great opportunity to address the challenges [2] [3] [4]. The objective is to develop and evaluate AI-driven platform that leverages the technology to provide smarter solution for efficient appointment booking [5] [6], rescheduling, cancelling, AI powered chatbot which is trained on Health Service Executive(HSE) Corpus to respond for quick medical queries and assisting patient needs. Also, providing an option for patients to wait in waiting list and notify the slot availability remainder [7] [8] for wait list patient instantly by optimizing the clinical time and reducing gaps in care due to cancellation or no-shows.

This modernizes the way to triage GP and patient engagement. Currently the application is for individual GP and their setoff clients for continuity of care. In future the platform also aims to provide the service Nation-wide to support multilingual voice-based NLP and onboard more GPs to launch a web platform for one time appointment, memberships request, waitlist signup and online GP visibility for rural areas nationwide. [9].

## II. RELATED WORK

The inclusion of AI into GP booking is essential for outpatient appointment and interaction, aiming to explore the human dependent traditional booking systems. This proposal, focusing on an AI-driven consultation platform

for GP appointment in Ireland, which is built upon existing research technologies while aiming specific gaps related to patient ease of experience, accessibility and efficiency.

#### A. AI-powered chatbots in GP booking

Studies explore AI chatbots for many healthcare applications for appointment booking. But use of chatbots for providing medical information, answering common queries, and offering general health support is very less. With advanced multimodal chatbot integrating text and voice, leveraging NLP and deep learning for enhanced interaction and access to a patient portal [10]. Also in [11], AI chatbot using NLP and ML for tasks such as appointment scheduling and symptom analysis, highlighting enhanced patient engagement. These work emphasise the AI chatbot increase communication and engagement but also lack integration still relying on manual bookings.

#### B. AI in appointment scheduling

Research depicts AI enhances scheduling and reduce waiting times. Clustering methods like K-Means, DBSCAN is utilised to prioritize patients by symptoms, then applied an MDP model [12]. Regression, time series, decision trees, and neural networks for medical scheduling [13]. These models improve resource use by strategically overbooking based on no-show likelihood, filling more slots via process redesign.

#### C. AI to predict patient no-shows

Prediction of patient no-shows reduces delays . In [14] models like Random Forest and Boosting performed best. In [15], 16 EHR features to predict no-shows through Random Forest, achieving 86% accuracy in risk-level categories, both of which include age and SMS reminders as top predictors. No-shows are predicted using ML together with important priorities like reminder actions or adjustment of the booking. Solutions (like reminders or overbooking) need to be implemented with caution. Also, some factors influencing the GP no-shows in online booking in Ireland are not clear.

### III. METHODOLOGY

A new smart platform for Irish GPs that integrates AI unlike other auto-cancellation solutions, our patients receive reminders 3 hours prior to the appointment to confirm, cancel, or reschedule appointments. Also offers the search of EIR-code for nearest specialists, and chat bot for medical query to establish holistic efficiency reducing administrative burdens.

#### A. System overview and architecture

The system encompasses different modules like mobile application, chatbot engine backed by REST based server and ML models for predication engine. Each of this component has functions to perform such as medical chatbot for scheduling, rescheduling, and cancellation of the appointment, search module to search specialist doctor based on the EIR code and availability, and ML model for prediction of no-show patients for missing appointments which enhance operational efficiency and patient engagement. The system architecture is shown in “Fig. 1” and technology stack in “Fig.2”.

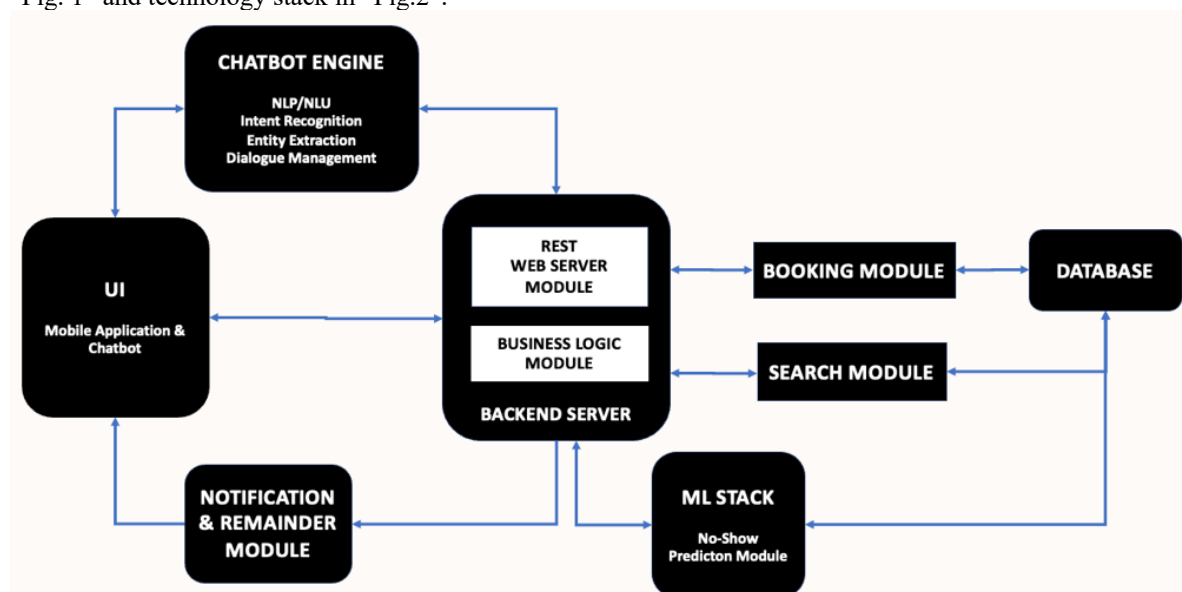


Fig. 1. System architecture

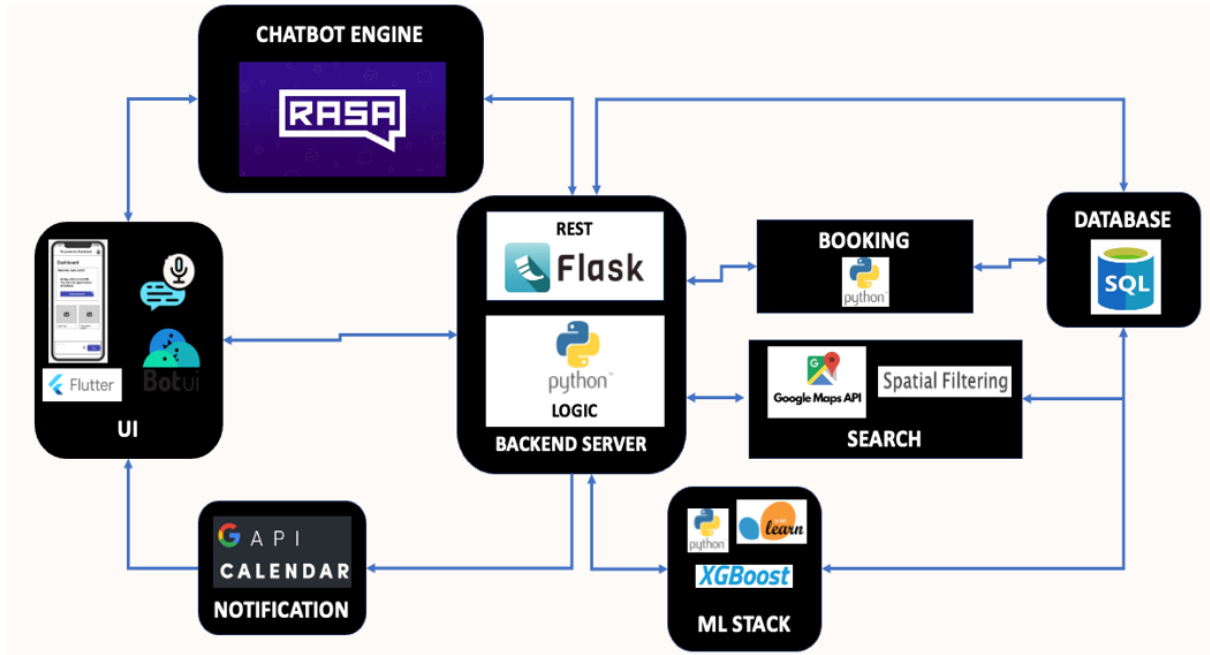


Fig. 2. Technology stack

Table 1 Technical components for AI-driven GP consultation.

| TABLE 1: TECHNICAL COMPONENTS |                                 |
|-------------------------------|---------------------------------|
| Modules                       | Technology                      |
| Mobile Application            | Flutter                         |
| Chatbot UI                    | BotUI                           |
| Notification System           | Google Calendar API + SMS/Email |

### B. System integration and sequence flow

A Flask-based REST API server handles backend logic in Python. A mobile app integrated with the RASA chatbot enables server communication and NLP-based text analysis. The booking module manages appointment scheduling. “Fig. 3”, shows search module flow and it uses Google Maps API and EIR codes to find specialists. SQL stores appointment and prediction related data. A Python-based ML stack would predict patient no-shows. Google Calendar API would send follow-up notifications. The model is periodically updated with new data.

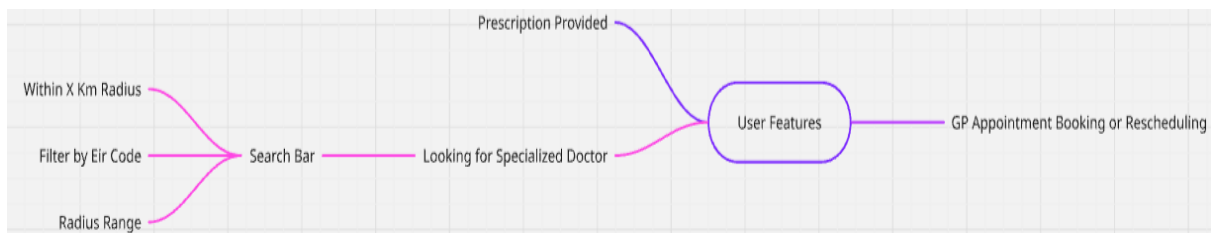


Fig. 3. Search module flow

“Fig. 4” showcases a sequence diagram of the process flow of booking a doctor’s appointment using AI-driven GP appointment system. The key components are User, GP Appointment System, Doctor, and Notification Service.

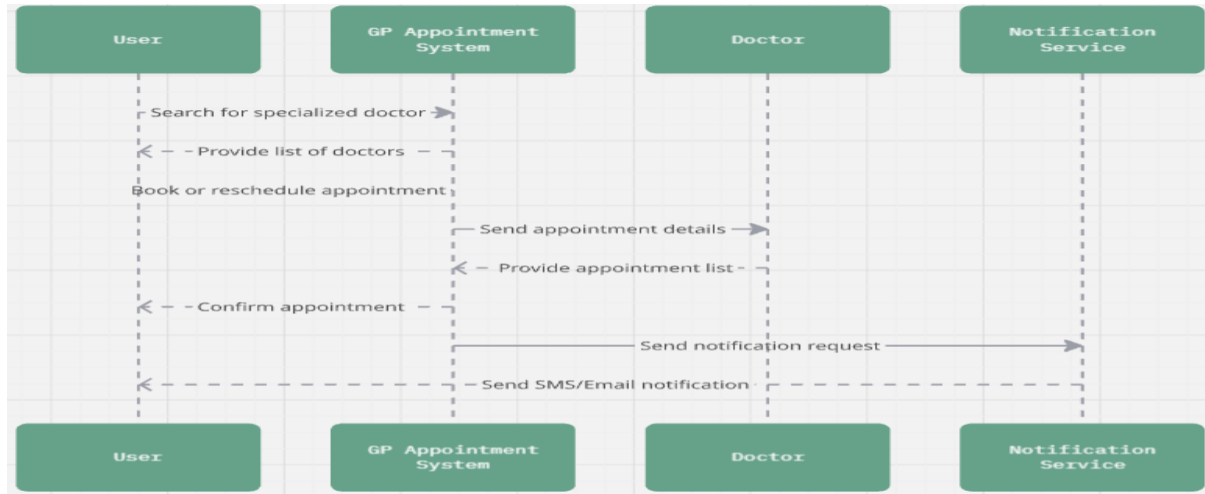


Fig. 4. Sequence flow

### C. Mobile application

The mobile application is the primary interface for patients to interact and is powered by Flutter on Android and IOS. This application serves booking, rescheduling and cancellation (for online/offline consultation) which is supported by RASA chat engine for voice/text interaction and google calendar API for real time notifications. “Fig. 5”, showcases the wireframe of the mobile application used in appointments.

Key functionalities include:

- EIR based on EIR code for best doctor using geo location.
- Real time alerts for waitlist and appointment confirmed patients.
- Chatbot medical queries.

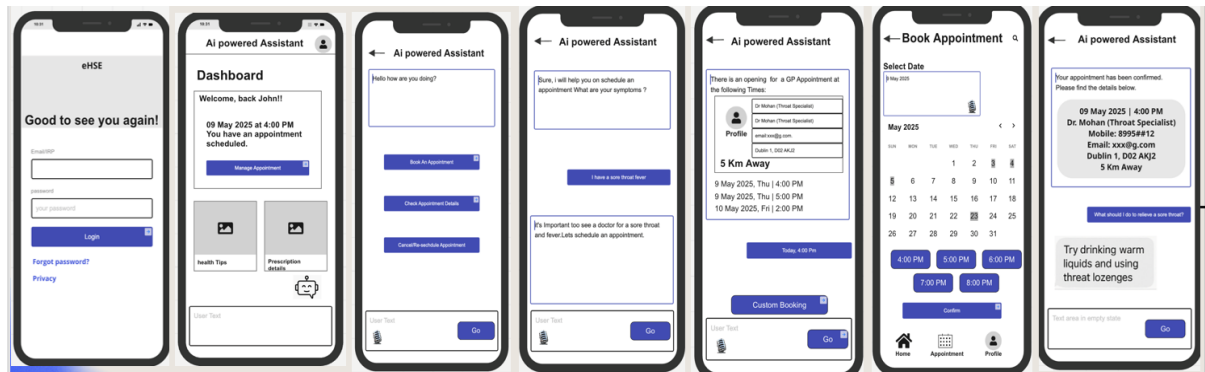


Fig. 5. Wireframe mock-up for mobile application

### D. Chatbot engine

The engine of the chatbot is built using RASA, which is properly customized to address medical terminologies as well as booking flows.

Customization involves:

- Intent Design & Entity Identification: Appointment management intents are designed and NLU pipelines is used to identify entities.
- Stories & Rules: RASA Core uses already defined stories and rules and lookup tables for dialogue management.
- Training Data: Chatbot is trained using domain specific phrases for improved medical jargon recognition.

### E. Backend module

Backend module consists of core REST server and business logics modules. Booking module and search modules are implemented using Python and Google Maps API and Spatial Filtering as highlighted in Table 2. SQL database is used to store structured data.

TABLE 2: SOFTWARE COMPONENTS

|                       |                                                 |
|-----------------------|-------------------------------------------------|
| REST Server           | Flask                                           |
| Server Business Logic | Python                                          |
| Booking Module        | Python                                          |
| Search Module         | Google Maps API and Spatial Filtering algorithm |
| Database              | SQL                                             |

#### F. ML implementation for no-show prediction

The dataset is derived from Kaggle source [16] contains information about medical appointments in Brazil such as patient details, appointment details, medical history, and behavioural factors. For Ireland, additional data is required like demographics(Eircode, urban/rural), behavioural features(no-show history, rescheduling), appointment metadata(type, speciality) and external factors(weather, holidays, travel distance).

##### a) Data Pre-processing

Missing data is handled by removing rows with irrelevant or lacking important features. Gender and age features can be computed using the mode and median method respectively. One-hot encoding for categorical features and label encoding for ordinal features. Min-Max scaling for numeric features is also done. Basic visualisations are done to see how the data are aligned with each other and before the implementation of algorithms dataset is split into 80/20 to maintain the class imbalance.

##### b) Model selection and training

Supervised classification models such as Logistic Regression, KNN, XGBoost, Decision Tree, Random Forest etc. has already been proven to provide exceptional and reliable accuracy [17]. Since the ML module majorly deals with classifying whether a patient will show up for appointment or not, it can be categorized as a classification problem and thus use the classification models. Reusing existing models aids in smooth onboarding & integration of proven ML stack into our application and thus send appropriate notification reminders.

Logistic Regression is used to predict the probability of a binary outcome through a logistic function for linearly separable data [18] as shown in “Fig. 5”.

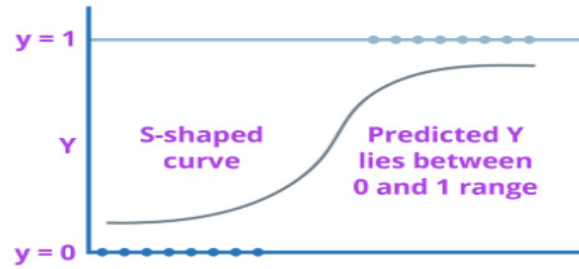


Fig. 5. Logistic Regression

Random Forest is group of decision trees where each of them casts a vote to the final decision avoiding overfitting[19] as shown in “Fig. 6”.

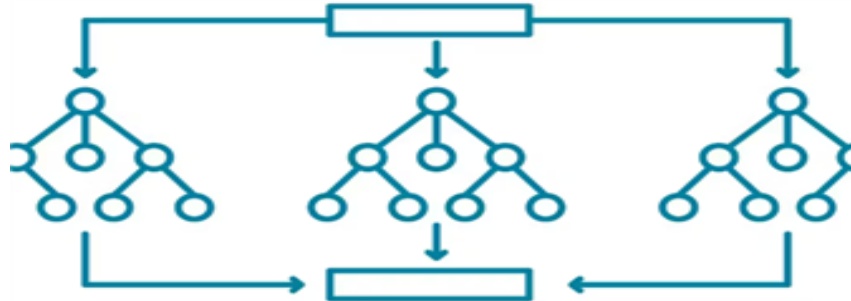


Fig. 6. Random Forest

XGBoost is gradient boosting which works well with structured/tabular and imbalanced data [20] as shown in “Fig. 7”.

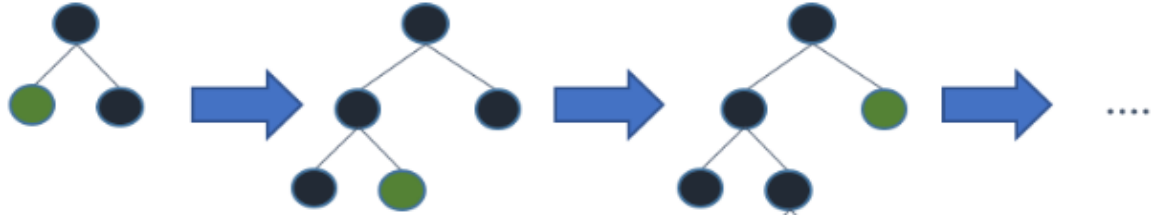


Fig. 7. XGBoost

#### IV. EVALUATION

##### A. Critical analysis of mobile application and chatbot engine

Flutter supports both IOS and Android eliminating the issue of cross-platform application design. RASA chatbot engine stands out by featuring development of custom scripts for inclusion of medical terms through custom Entity and Intents in YAML file.

##### B. Critical analysis of backend

Python includes all necessary tools, modules, libraries and drivers to setup web servers using Flask, build business logic and connect to databases such as SQL and ML module.

##### C. Critical Analysis of AI Techniques

Supervised Learning is best suited for structured tabular data [21]. Ensemble Methods such as XGBoost and Random Forest will improve result by combining weak learner network. Neural Networks can be used but requires more data for optimal performance.

| Model                      | Accuracy     | Precision   | Recall      | F1-Score    | AUC-ROC     |
|----------------------------|--------------|-------------|-------------|-------------|-------------|
| Logistic Regression        | 0.796        | 0.82        | 0.75        | 0.78        | 0.81        |
| Random Forest              | 0.801        | 0.83        | 0.77        | 0.80        | 0.84        |
| Decision Tree              | 0.72         | 0.32        | 0.32        | 0.32        | 0.32        |
| XGBoost                    | <b>0.804</b> | <b>0.85</b> | <b>0.78</b> | <b>0.81</b> | <b>0.86</b> |
| Voting Classifier          | 0.804        | 0.84        | 0.77        | 0.80        | 0.85        |
| MLP Classifier             | 0.787        | 0.81        | 0.74        | 0.77        | 0.80        |
| Gradient Boosting          | 0.802        | 0.83        | 0.76        | 0.79        | 0.83        |
| K-Nearest Neighbours (KNN) | 0.770        | 0.79        | 0.72        | 0.75        | 0.78        |

Fig.8. Model performance analysis

| Metric                | Purpose                                         | Justification                                                                                                            |
|-----------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Accuracy              | Measures overall correctness.                   | Standard metric, but not reliable for imbalanced data.                                                                   |
| Precision             | Minimizes false positives.                      | Ensures reminders are sent only to patients that have more probability of missing appointments.                          |
| Recall (Sensitivity)  | Maximizes true positives.                       | Critical to avoid missing high-risk cases. Ensures frequent patients who don't show up for appointment are not left out. |
| F1-Score              | Harmonic mean of precision and recall.          | Balances between false positives and false negatives.                                                                    |
| AUC-ROC               | Evaluates model discrimination or bias ability. | Works well for imbalanced classification.                                                                                |
| Confusion Matrix      | Visualizes TP, FP, FN, TN.                      | Helps interpret errors of model.                                                                                         |
| No-Show Reduction (%) | Measures real-world impact.                     | Used to send notification based on the threshold percentage.                                                             |

Fig 9. Performance metrics

“Fig. 8” and “Fig. 9” highlights the performance metrics along with the model performance analysis where XGBoost performed well, compared to other models as per the evaluation metrics. Bar chart showcased in “Fig. 10” is comparing the algorithms with accuracy values.

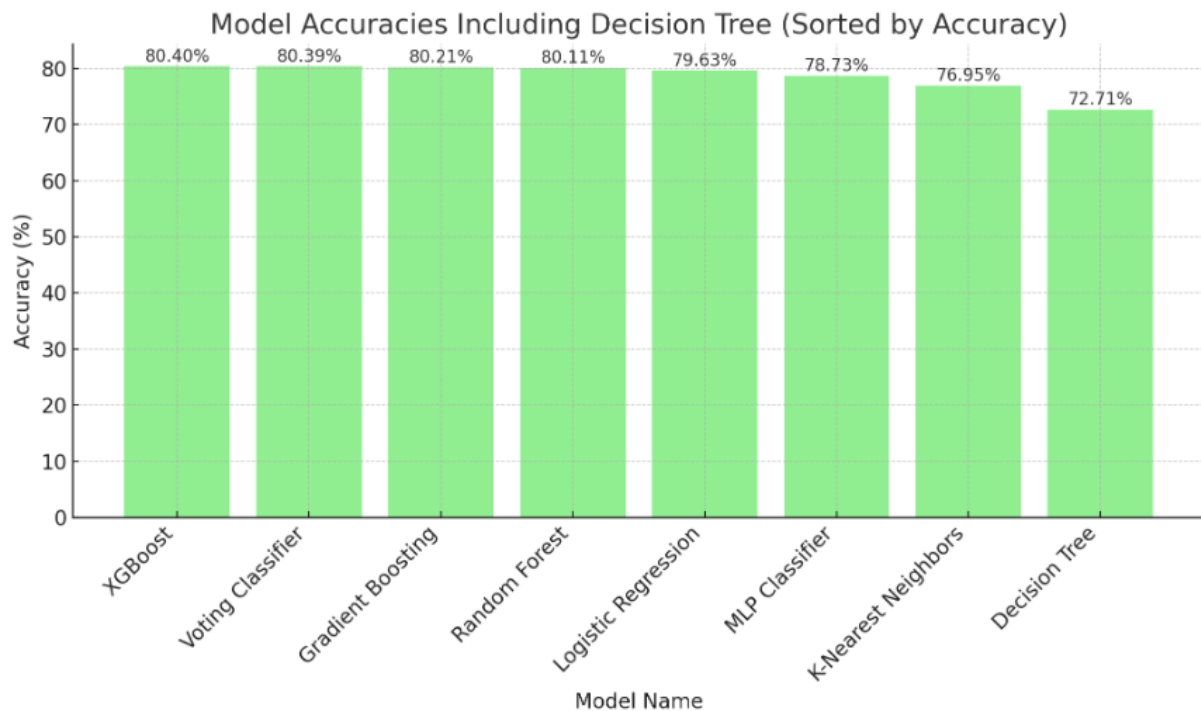


Fig.10. Model accuracy

## V. CONCLUSION & FUTURE WORK

The project demonstrates the potential of AI in enhancing GP booking system in Ireland. By using ML and NLP, we have proposed a unique platform in this report that organises appointment scheduling, minimizes no-shows, and amplifies patient engagement. Key modules such as AI chatbot for real-time support, reminders for no-show and EIR based search for specialist access modules collaboratively offers strong and scalable solution.

The usage of XGBoost showed best performance over the metrics for the real-time complex scenario. There is a potential to expand platform coverage all over Ireland with multilingual voice-enabled NLP booking. Personalization can also be improvised through AI-remedy suggestions and integrate image-based diagnosis. Additionally, we can aim for a nationwide centralised platform for GPs to evolve as a national hub for GP service to onboard more practitioners and enable right matches to the right specialists quickly. Ethical AI governance and GDPR compliance will remain as core principles, making sure that data privacy and transparency are secured. The no-show prediction model is aimed to adapt for continuous learning, trends, providing a patient-centric and tech-enabled healthcare system in Ireland and EU region.

## REFERENCES

- [1] T. Journal, "Ireland already has some medical deserts - and it's been getting worse," *The Journal*, 2025.
- [2] A. K. Mishra, N. Singh, and A. Nayak, "AI-Driven Appointment System for Primary Health Centers," in *\*Proc. 2024 Int. Conf. on Convergence of Innovative Technologies (CONIT)\**, 2024, pp. 1–6. doi: 10.1109/CONIT61985.2024.10626795..
- [3] P. Gupta and R. Sharma, "Natural Language Based Doctor Appointment Booking Chatbot," in *\*Proc. 2022 7th Int. Conf. on Communication and Electronics Systems (ICCES)\**, 2022, doi: 10.1109/I2CT54291.2022.982500.
- [4] K. Mahato, R. Mishra, A. Sen, A. Raj, and S. Kumar, "Smart Scheduling System for Healthcare: An AI-based Solution," in *\*Proc. 2024 IEEE MITADTSOciCon\**, 2024, doi: 10.1109/MITADTSOciCon60330.2024.10575404.
- [5] Y. Liang and L. Zhao, "Intelligent Hospital Appointment System Based on Health Data Bank," *\*Elsevier\**, 2019. [Online]. [Accessed: Apr. 02, 2025].
- [6] G. Fan, Z. Deng, Q. Ye, and B. Wang, "Machine learning-based prediction models for patients' no-show in online outpatient appointments," *\*Int. J. Hum.-Comput. Interact.\**, vol. 38, no. 4, pp. 1–10, 2022, doi: 10.1080/10447318.2022.2089085.
- [7] L. A. Silva, B. M. Bermejo Gil, F. P. Robledo, I. M. Pires, V. R. Q. Leithardt, and A. M. F. da Rocha, "Integrating AI and telemedicine in maternal-infant care: An innovative approach for personalized healthcare," [Accessed: Apr. 3, 2025].
- [8] A. M. and R. U. R. Kadapal, "Leveraging AI-Driven Analytics in Product Management for Enhanced Business Decision-Making," *Library Progress International*, vol. 44, no. 3, pp. 22136–22144, 2024.
- [9] E. M. and S. Connolly, "Remote Consultations in General Practice in Ireland: Who is missing out?," *Economic and Social Research Institute (ESRI)*, 2025.
- [10] R. C. Hegde, S. Garimella, "Advancing Healthcare Accessibility: Development of an AI-Driven Multimodal Chatbot," in *Proc. 2024 4th Int. Conf. on Intelligent Technologies (CONIT)*, 2024.
- [11] S. D. G., et al., "Chat-bot Health-Care Application using Artificial Intelligence," in *Proc. 2024 MIT Art, Design and Technology School of Computing International Conference (MITADTSOciCon)*, 2024.
- [12] N. Yousefi, F. Hasankhani, M. Kiani, and N. Yousefi, "Appointment scheduling model in healthcare using clustering algorithms," 2023.
- [13] G. L. N.-N., F. T. E. C. Valenzuela-Núñez, and F. Troncoso-Espinosa, "Smart Medical Appointment Scheduling: Optimization, Machine Learning, and Overbooking to Enhance Resource Utilization," in *Proc. IEEE*, 2024.
- [14] G. Fan, Z. Deng, Q. Ye, and B. Wang, "Machine learning-based prediction models for patients' no-show in online outpatient appointments," *Healthcare Analytics*, vol. 1, 2021.
- [15] M. C. E. Simsekler, M. A. Omar, K. M. Toffaha, and I. ElKebbi, "Predicting patient no-shows using machine learning: A comprehensive review and future research agenda," *Intelligent Medicine*, vol. 2025, pp. 1-10, 2025.
- [16] F. Krimmelbein, "Medical Appointment No Shows," August 2017.
- [17] G. Fan, Z. Deng, Q. Ye, and B. Wang, "Machine learning-based prediction models for patients no-show in online outpatient appointments," *KeAi Data Science and Management*, vol. 2021, pp. 1-10, 2021.
- [18] L. H. Salazar, A. Fernandes, R. Dazzi, N. Garcia, V. R. Q. Leithardt, "Using different models of machine learning to predict attendance at medical appointments," *J. Inf. Syst. Eng. Manag.*, vol. 2020, pp. 1-10, 2020.
- [19] L. Breiman, "Random Forests," *Machine Learning*, vol. 45, 2001.
- [20] T. C. Carlos Guestrin, "XGBoost: A Scalable Tree Boosting System," in *ResearchGate*, 2016.
- [21] C. Deina, F. S. Fogliatto, G. J. C. da Silveira, and M. J. Anzanello, "Decision analysis framework for predicting no-shows to appointments using machine learning algorithms," *BMC Health Serv. Res.*, vol. 2024, pp. 1-10, 2024.